

What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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TIME-DOMAIN ASTROPHYSICS

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Introduction

Contacts

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Main Goal of the course: Have fun!

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Figure1: Near-IR observation of a high density environment.



Figure2: Hot Spots in Fig 1, with regression showing the outlier.



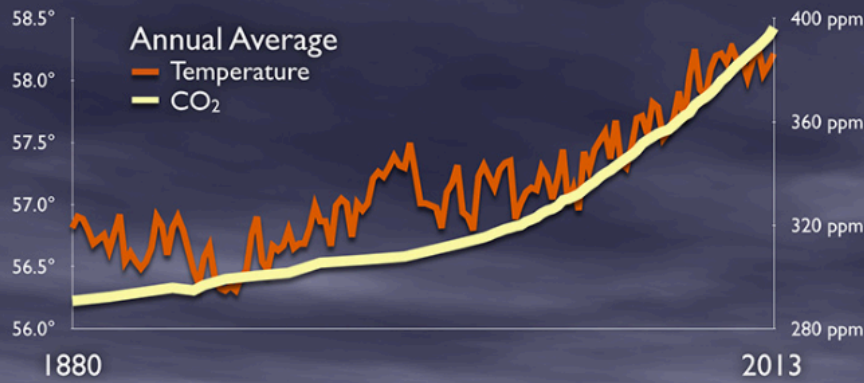
- This is serious. Specialized academic courses, in a manner of speaking, should be enjoyed...!

Time-Series are ubiquitous

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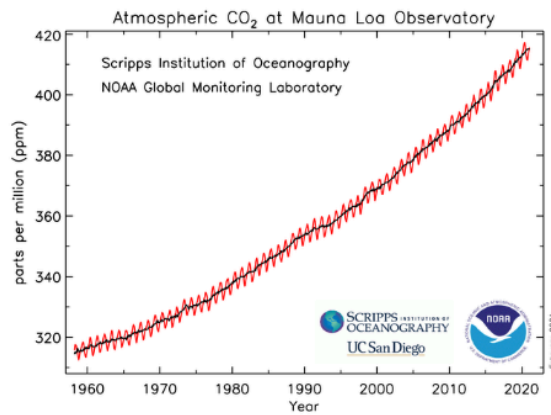
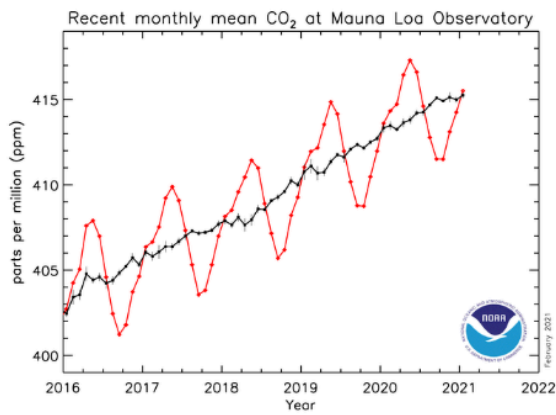
- Anytime we have a measurement repeated multiple times we have a time-series.

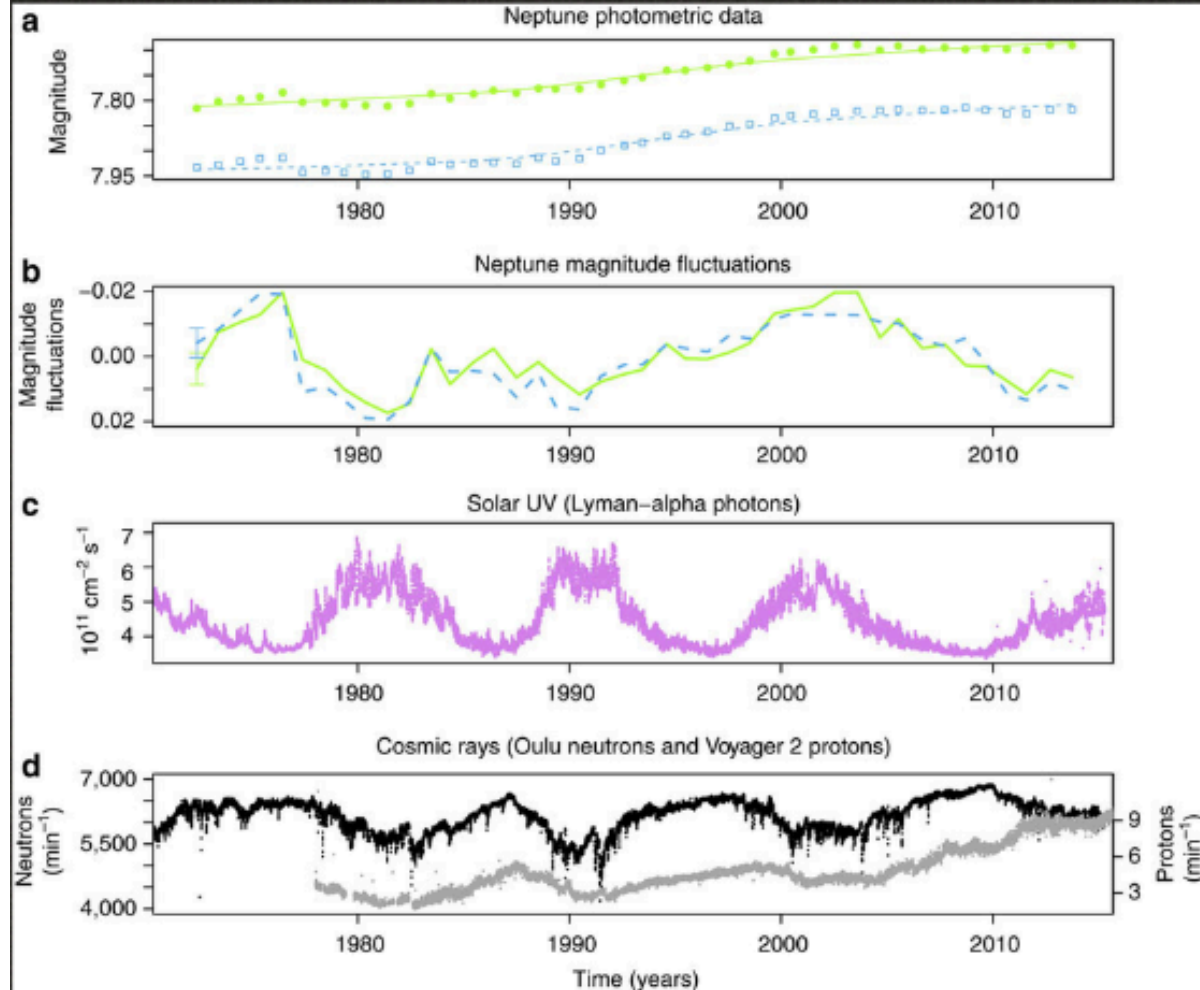
Global Temperature and CO₂



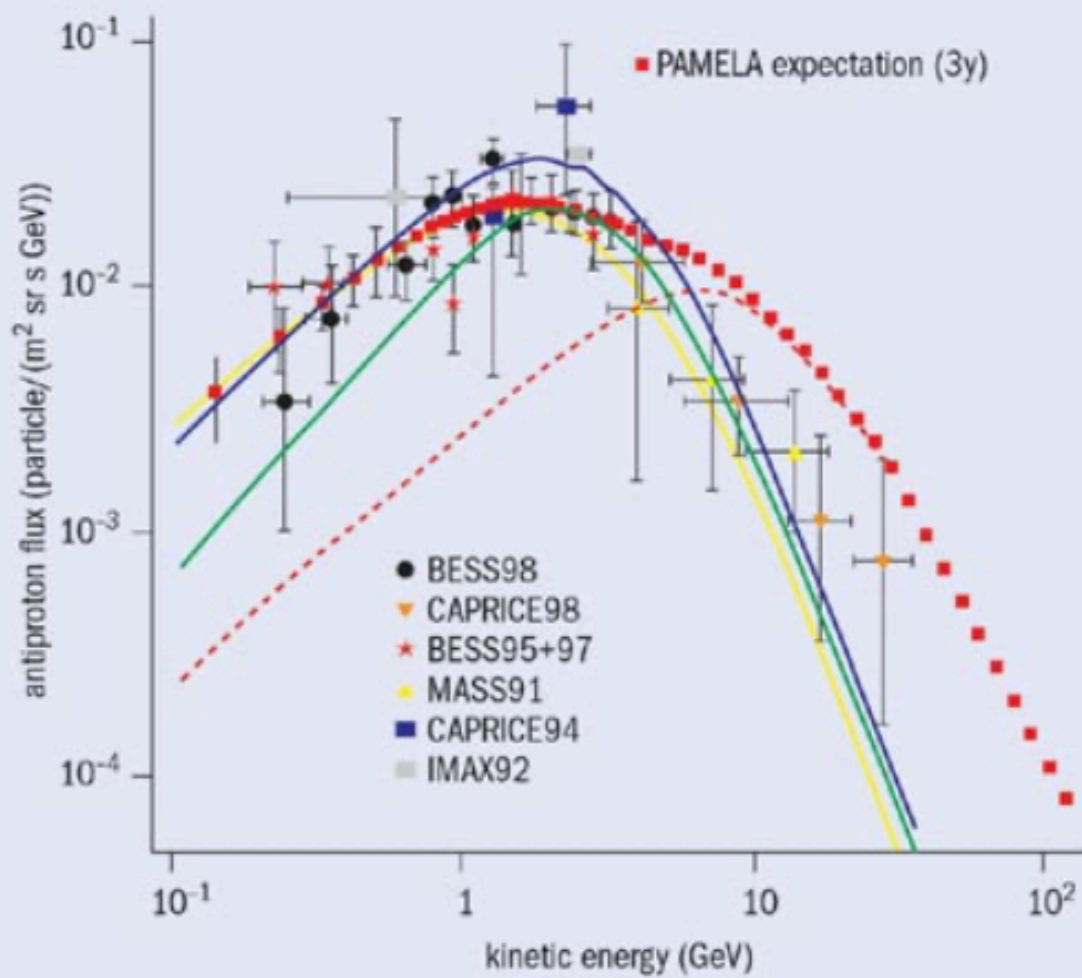
Source: National Climate Assessment 2014

CLIMATE CENTRAL





- As a matter of fact, a time-series does not need to have "time" as index!





Temptative program (it may change...)

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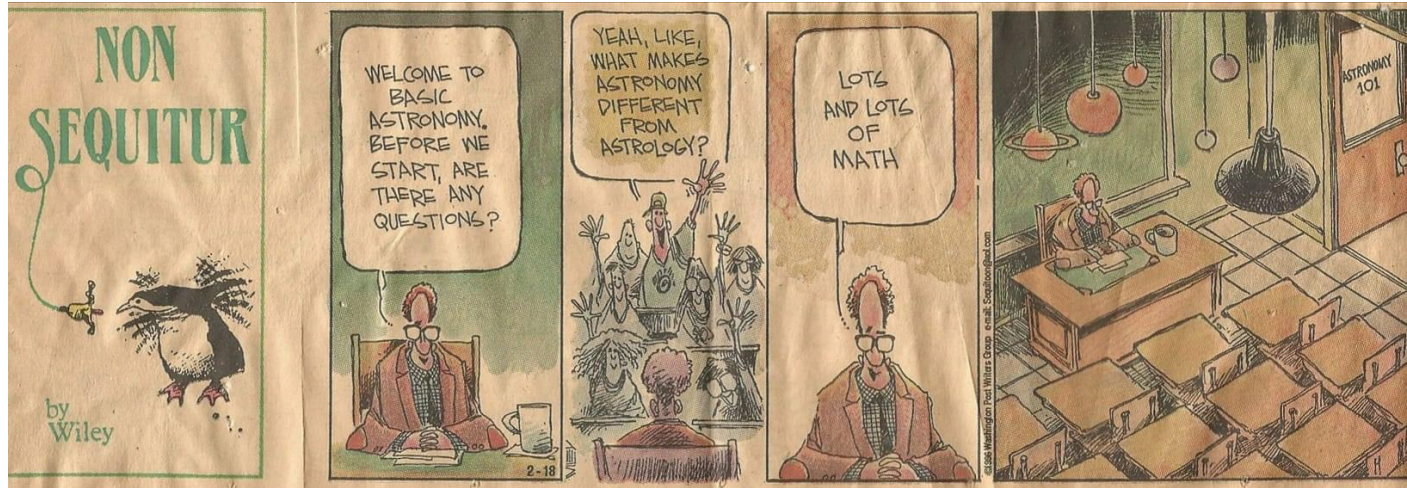
1. Introduction
2. Statistics reminder - part I
3. Statistics reminder - part II
4. Spectral analysis - part I
5. Spectral analysis - part II
6. Science cases: Sunspots Number - X-ray Binaries
7. Irregularly sampled time series - part I
8. Irregularly sampled time series - part II
9. Science Cases - Variable Stars
10. Time domain analysis - part I
11. Time domain analysis - part II
12. Science Cases - AGN and blazars
13. Wavelet analysis
14. Science case about climatology
15. Time of arrival analysis and paleo-climatology science case
16. Science case: FRBs
17. Non-parametric methods - part I
18. Non-parametric methods - part II
19. Science case: GRBs
20. Singular spectrum analysis - part I
21. Singular spectrum analysis - part II
22. Gaussian processes - part I
23. Gaussian processes - part II
24. Astrostatistics: final considerations

How is the course managed?

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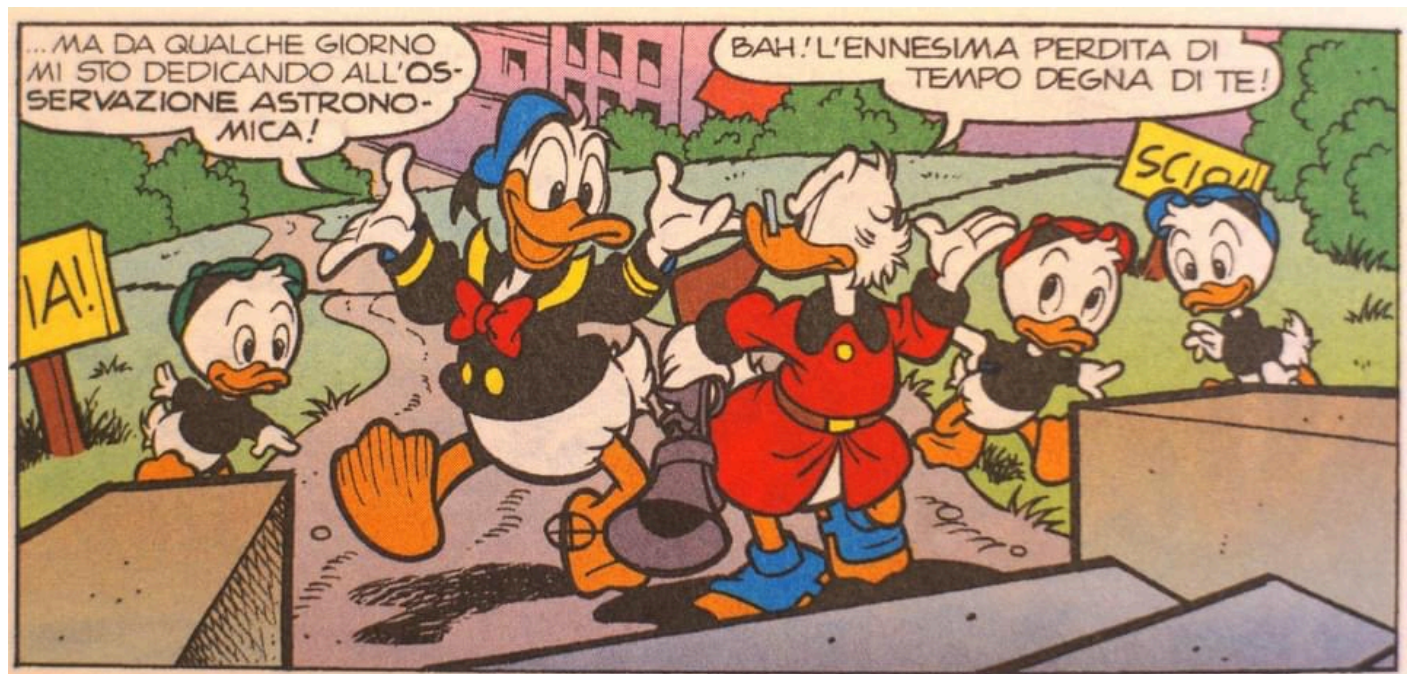
Frontal lectures

- These are the traditional university lectures.
- Although this increases the organizational complexity substantially, I am available to stream and record my lectures, if needed.
- There are contraindications. As a matter of fact, this is one of few cases where a remote access is not even close as effective as being in presence.



Real research life examples...

- Scientists working in the field will deliver "didactic lectures", allowing one to see most of ideas developed during the course applied in a real research environment.



(Optional) papers to deepen our knowledge...

- Most of the topics discussed during the course can be investigated thoroughly and papers from astrophysical (mainly) literature are presented for particularly concerned readers.



Question time

- The course is divided in several main sections. At the end of each of them, some time will be devoted to open discussions and questions.



Lectures from specialists in the field

- Together with regular lectures, a few specialists in the field, i.e. scientist carrying out researches by time-domain tools and techniques, are invited to describe their works.



Language

- According to university guidelines, lectures will be delivered in English. Of course, a fair evaluation of the context might ask some flexibility.



Statistical framework

- During this course we are going to work in a Bayesian framework.
- Bayesian statistics is an approach to inferential statistics based on Bayes' theorem, where available knowledge about parameters in a statistical model is updated with the information in observed data. The background knowledge is expressed as a prior distribution and combined with observational data in the form of a likelihood function to determine the posterior distribution. The posterior can also be used for making predictions about future events.
- Nevertheless, we are not dogmatic and mentions or applications based on familiar "frequentist" approaches are presented and discussed, when we deem it opportune.



Programming languages

- Most of the examples we are going to analyze during the course are based on some sort of computer analysis.
- Python is *de-facto* the standard language in data science.
 - Yet, while this language is definitely truly amazing, well designed and worth mastering, for the specific needs of scientific computing there are alternatives of growing popularity.
- We therefore provide examples written in Julia, and encourage the students to get some confidence with this programming language too.
- Notebooks are written by the markdown language, a simple language integrating features of the HTML and LaTeX languages.
- We use here a specific flavours of notebook known as Pluto notebooks. These notebooks offer a high level of interactivity and several more advanced features. However, any other option is perfectly fine, e.g., jupyter, Visual Studio, etc.



- There are plenty of good tutorial about the Julia language. We can suggest trying this introduction for scientists available online, e.g. Julia data science. Else, this introduction, in spite of being, formally, designed for economists, is of particular interest and worth considering for any computer scientist.

Warning! The course is not only for astrophysicists!

- It is indeed part of the set of courses for future astrophysicists. Nevertheless, almost nothing we are going to discuss is truly only for astrophysics. In reality, several applications and ideas are taken from other fields, i.e. economics, social sciences, climatology, etc.

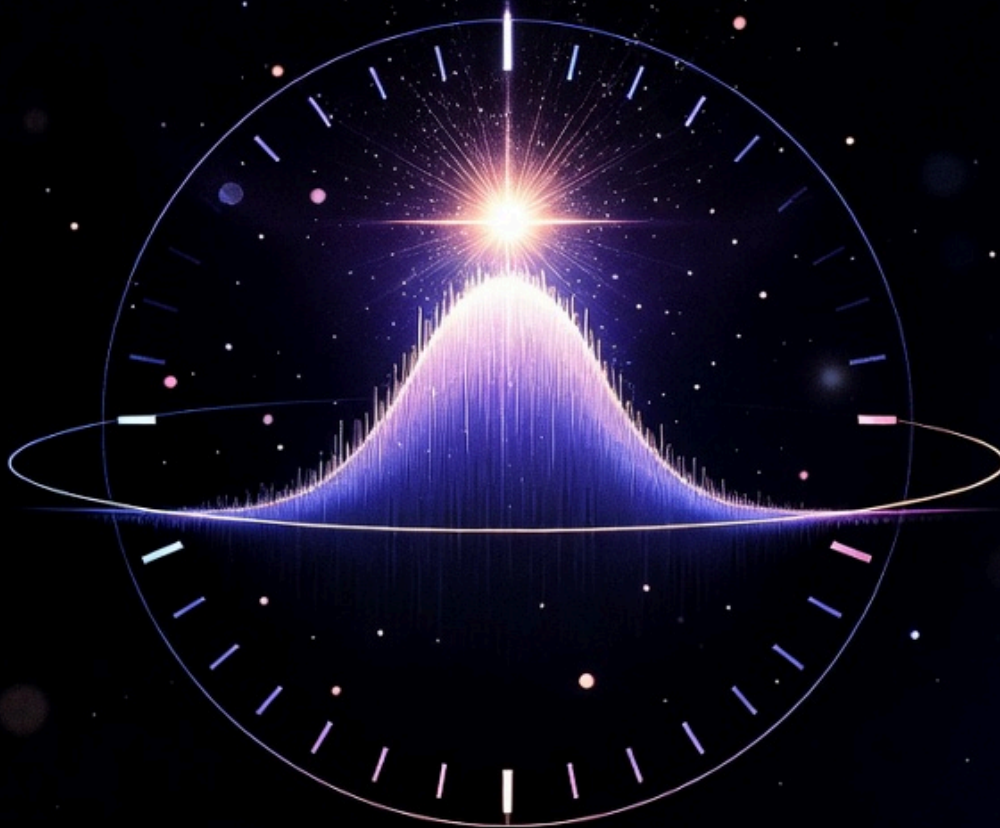


Final assessment

- The final examination is an oral one.
- *Students* must interact with the teacher in advance of the examination and a science case obtained by the modern literature will be selected.
- The *student* will be asked to properly describe the main formal aspects of the study and discuss critically the reliability and limits of the presented results.

As a general rule, in order to take the exam, attending $\sim 50\%$ of the lectures is required.

TIME-DOMAIN ASTROPHYSICS



TIME-DOMAIN
ASTROPHYSICS

- For most of the users, a web site reporting all the material shown during the lectures is probably the preferred and simpler solution. The course is available online at: [website](#)

The site enables to run interactively all the notebooks with examples and/or exercises discussed during the course!

Github repository

- Pluto notebooks, papers, etc. are also available on [github](#)
- Check the repository frequently since is (rather often) updated during the course.



How to clone the repository:

- It is a rather "nerd" solution!
- Open a terminal and move to a directory where material will be stored.

```
cd mydir
git clone https://github.com/stefanocovino/Time-Domain-Astrophysics.git
cd TimeDomainAstrophysics
git pull
```

- This clones the whole tree (i.e. the course material, about 3GB).
- In order to start reading the notebooks one can execute, i.e.:

```
cd TimeDomainAstrophysics
julia -e e 'using Pluto; Pluto.run(notebook="Course.jl")'
```

- Repeating frequently the last command (`git pull`) you will always have the tree fully updated and you notebooks, data, papers, etc. ready to be used on your computer.

Cloning the repository is only needed if one wants to get and interact with the full notebooks, with code, data, etc.

- Single files, papers, etc. can be downloaded browsing the repository or directly by the links included in the notebooks.

Course calendar and remote connection

The course calendar, notes, advised, topics discussed during a given lecture, and remote connection details can be found at this [url](#).

Relaxing time(-series...)



Reference & Material

- The course is based on published scientific papers distributed by the teacher before any main topic is addressed.
- Science cases are based on actual scientific papers as well.
- Slides prepared by the teacher will also be distributed.
 - A general introductory text to time series analysis as: [“Introduction to Time Series and Forecasting”, by P.J. Brockwell and R.A Davis](#) might be useful. However, any other analogous text easily obtainable by the student will be fine as well.
- Two textbooks more strictly related to the topics discussed during the course mainly, but not only, for astrophysical applications are:
 - [“Modern Statistical Methods for Astronomy”, by E.D. Feigelson and G.J. Babu](#)

- [“Statistics, data Mining and Machine Learning in Astronomy”, by Ivezić et al.](#)

Further Material

Papers for examining more closely some of the discussed topics.

- [Voughan et al. \(2013\) - "Random Time Series in Astronomy"](#)
- [Storopoli et al. \(2021\) - "Julia Data Science"](#)

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Course Summary	Statistics Reminder	Course Summary
html	Course Summary	Statistics Reminder	Course Summary

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Notebook v1.0.2 - 27 April 2026